

Question 1 Answer:

A **copy constructor** is a constructor that creates a new object by copying the values of fields from another object of the same class or struct. It's typically used when you need a **duplicate object with the same state** but as a separate instance, avoiding shared references that could lead to unintended side effects.

Question 2 Answer:

An **indexer** in C# allows objects to be accessed using **array-like syntax** (obj[index]). It's used when a class or struct logically represents a collection of data and you want intuitive access to its elements without exposing the internal storage mechanism.

Business use cases for indexers:

- **Product Catalog** – Access products by index or product code.
- **Employee Directory** – Retrieve employee details by employee ID.
- **Configuration Settings** – Access settings using string keys.
- **Matrix or Grid Data** – Access rows and columns in numerical computations.